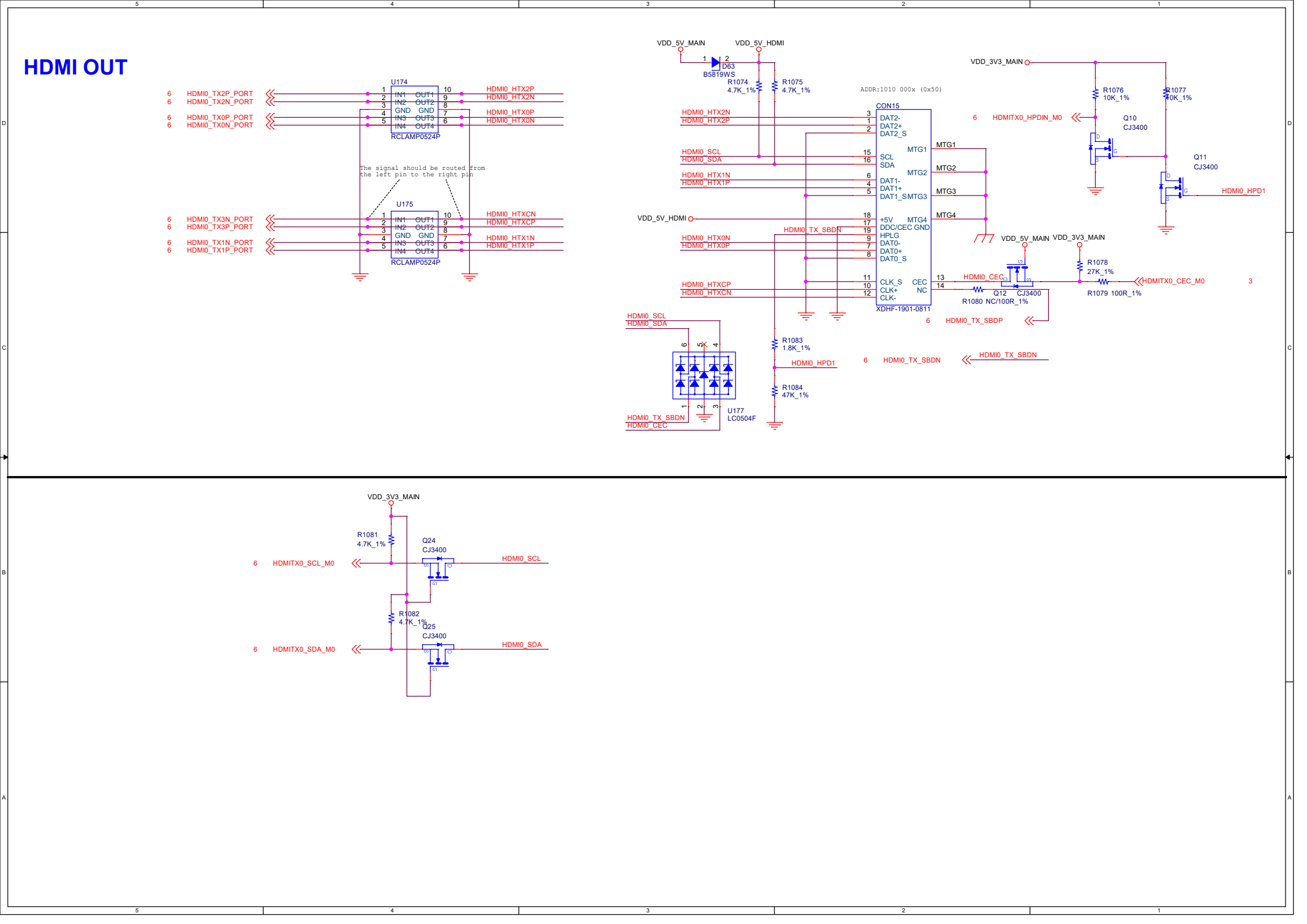
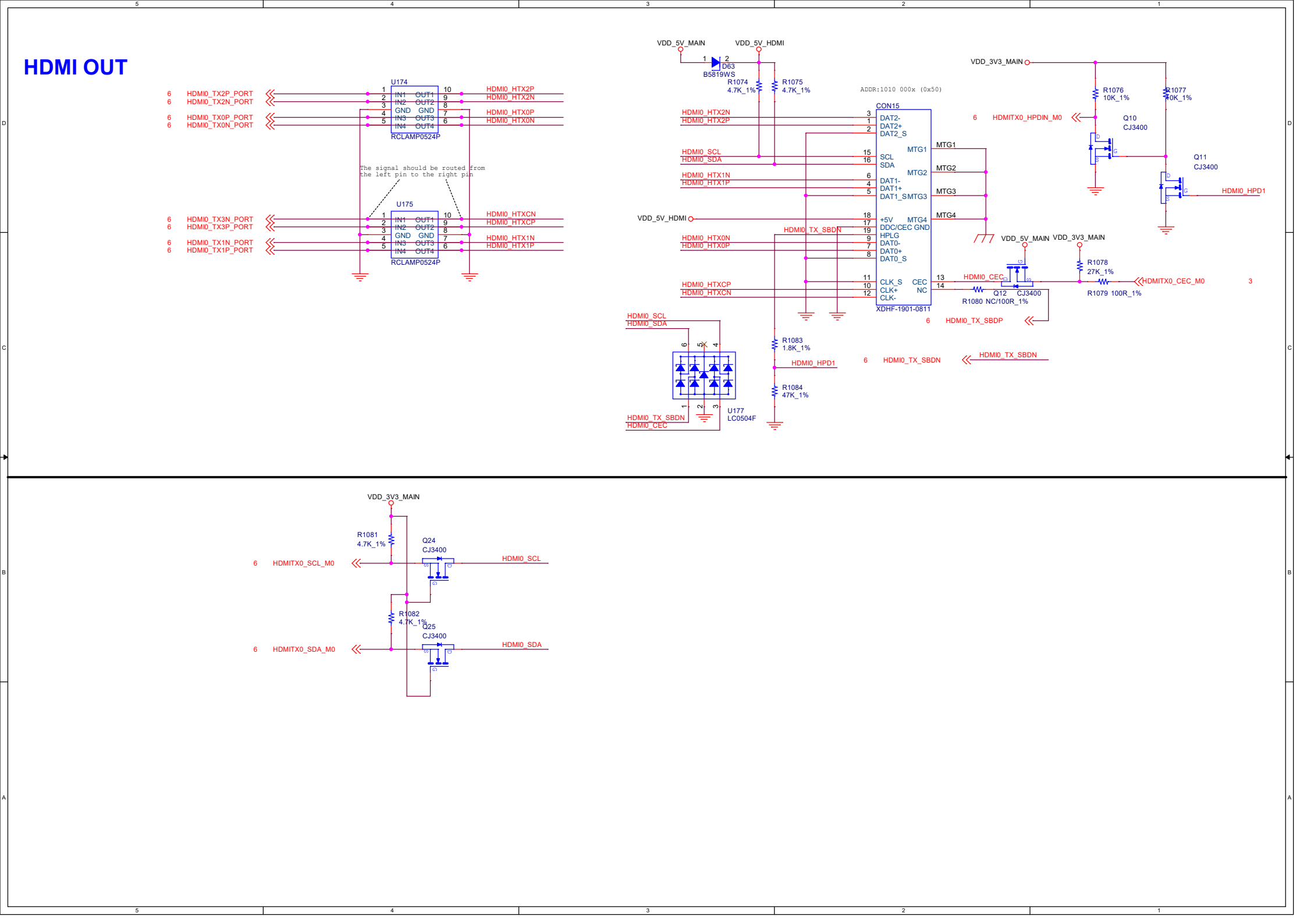
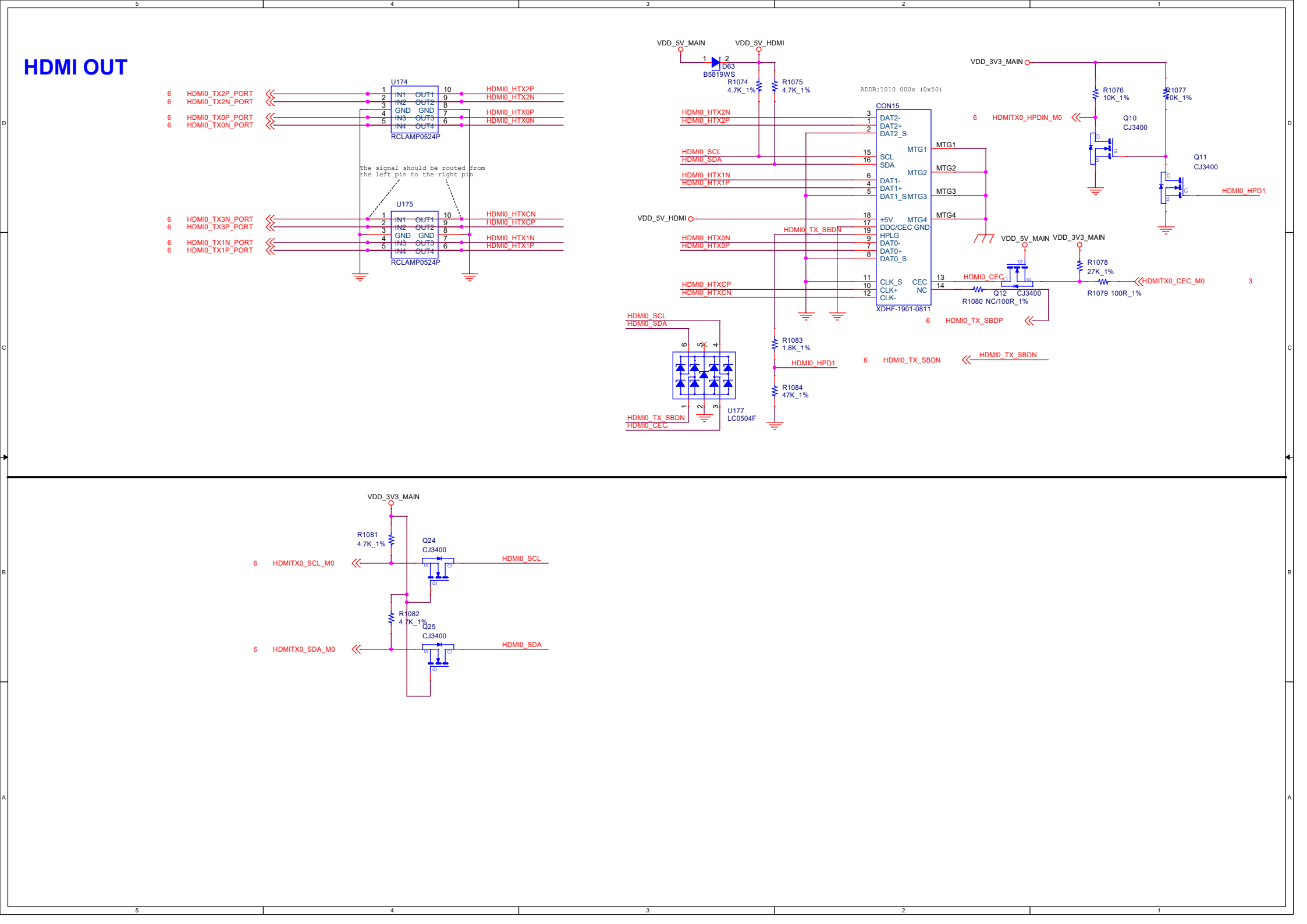
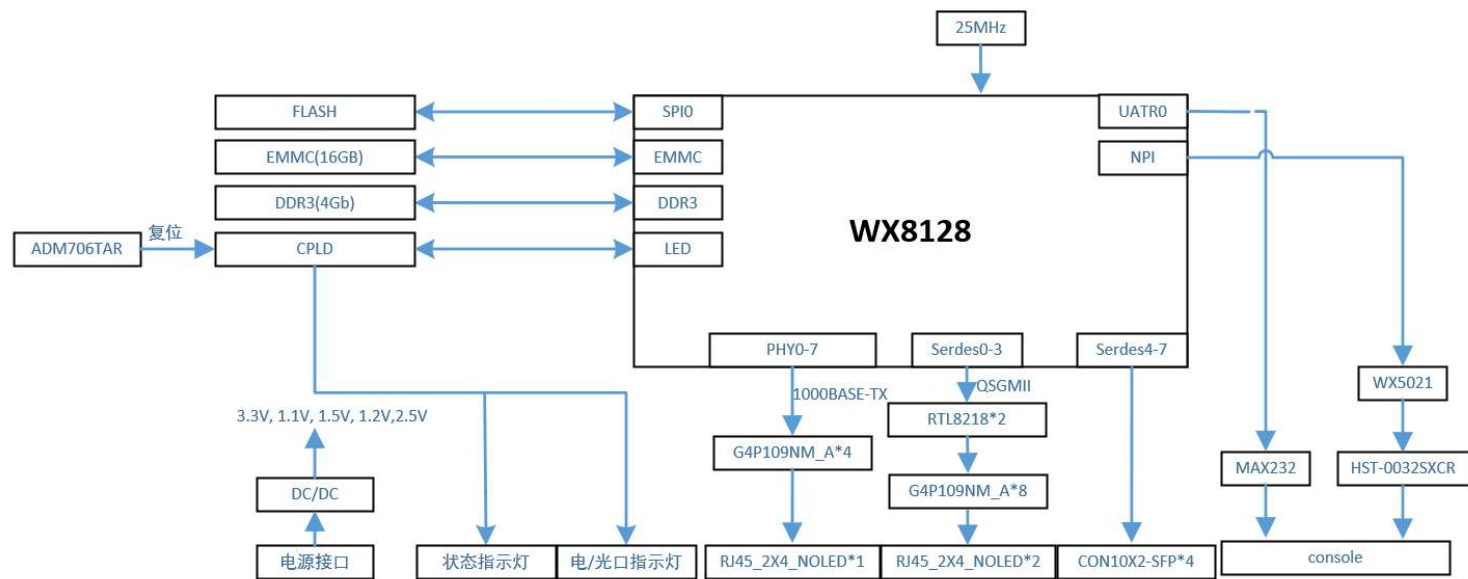
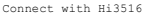


[illegible]



北京物芯科技有限责任公司			
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Friday, March 28, 2025		PCB对应版本:	V1.0
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图纸尺寸:	C	文档编码:	V1.0
日期时间:	<Date>	页次:	1 of 19



SOM CONNECTOR CON0A/C



14,15,16 BOOT_SARADCIN0
14,15,16 PWRON_L
14,15,16 PMIC_EXT_EN_OUT
14,15,16 GPIO2_C1/MIPI_CAM2_RESET_L
14,15,16 GPIO2_C2/MIPI_CAM2_RESET_L

21 MIPI_CS11_RX_D3P
21 MIPI_CS11_RX_D3N
21 MIPI_CS11_RX_D2P
21 MIPI_CS11_RX_D2N
21 MIPI_CS11_RX_CLKOP
21 MIPI_CS11_RX_CLKON
21 MIPI_CS11_RX_D1P
21 MIPI_CS11_RX_D1N
21 MIPI_CS11_RX_D0P
21 MIPI_CS11_RX_D0N

19,21 GPIO1_B6/MIPI_CAM1_CLKOUT
21 GPIO1_B7/MIPI_CAM2_CLKOUT

21 GPIO2_A6/SPI_D0_M1
21 GPIO2_A7/SPI_D1_M1
21 GPIO2_B0/SPI_D2_M1
21 GPIO2_B1/SPI_D3_M1
21 GPIO2_B2
21 GPIO2_B3/SPI_CLK_M1
21 GPIO2_B4/SPI_CS0_M1
21 GPIO2_B5
21 GPIO2_B6
21 GPIO2_B7

9 RESET_L
9 VDC_MODE
19 UART2_TX_M0_DEBUG
9 UART2_RX_M0_DEBUG

9 LED3_25GLAN_25GLED
21 LED2_25GLAN_1GLED
21 P25G_MDIP0
10 P25G_MDIP0
21 P25G_MDIP1
10 P25G_MDIP1
21 P25G_MDIP2
10 P25G_MDIP2
21 P25G_MDIP3
10 P25G_MDIP3

14,15,16 RK3588_RUN
14,15,16 RK3588_ST1
15 GPIO4_A2/M2_D_PERST_L
14,15,16 RK3588_ST2

15 GPIO4_B0/USB3_TYPEC1_PWREN
16 GPIO4_B3/M2_C_PERST_L
15 GPIO4_B4/M2_B_PERST_L
15 GPIO4_B5/M2_A_CLKREQ_L
15 GPIO4_B6/M2_A_PERST_L
16 GPIO1_A0/UART6_RX_M1
16 GPIO1_A1/UART6_TX_M1

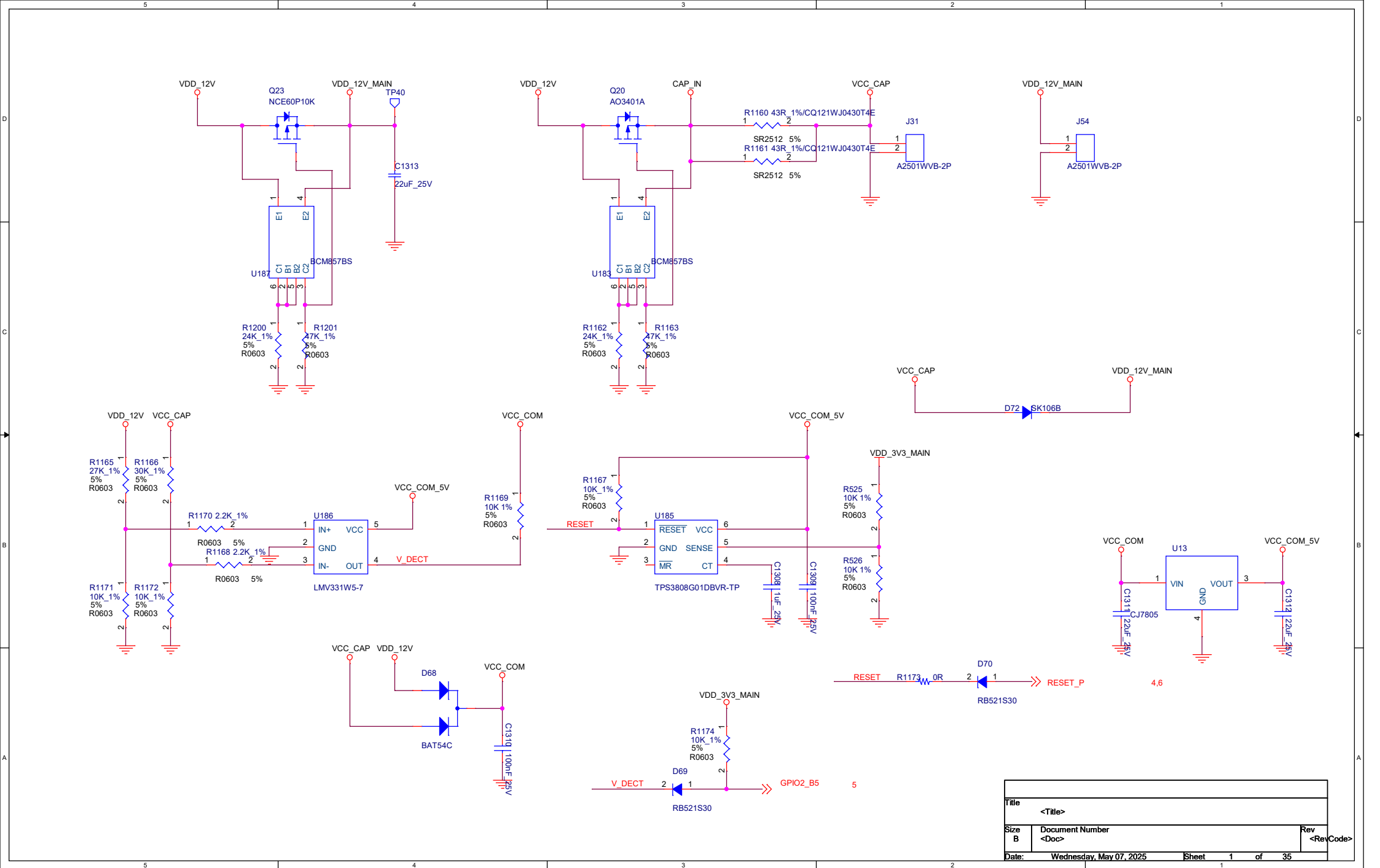
16 GPIO1_A4/USB2_PWREN
16 GPIO1_A7
15 GPIO3_C0/TP_INT_L
16 GPIO3_C1/TP_RST_L
15 GPIO3_C3/PWM14_M0
16 GPIO3_C3/PWM15_IR_M0
15 HDMITX_HPDIN_M0
16 HDMITX_HPDIN_M0
15 GPIO1_D5/HDMI_RX_DET_L

VDD_3V3_MAIN0
R511 10K_1%

VDD_3V3_MAIN0
R1117 10K_1% NC
R1118 10K_1%

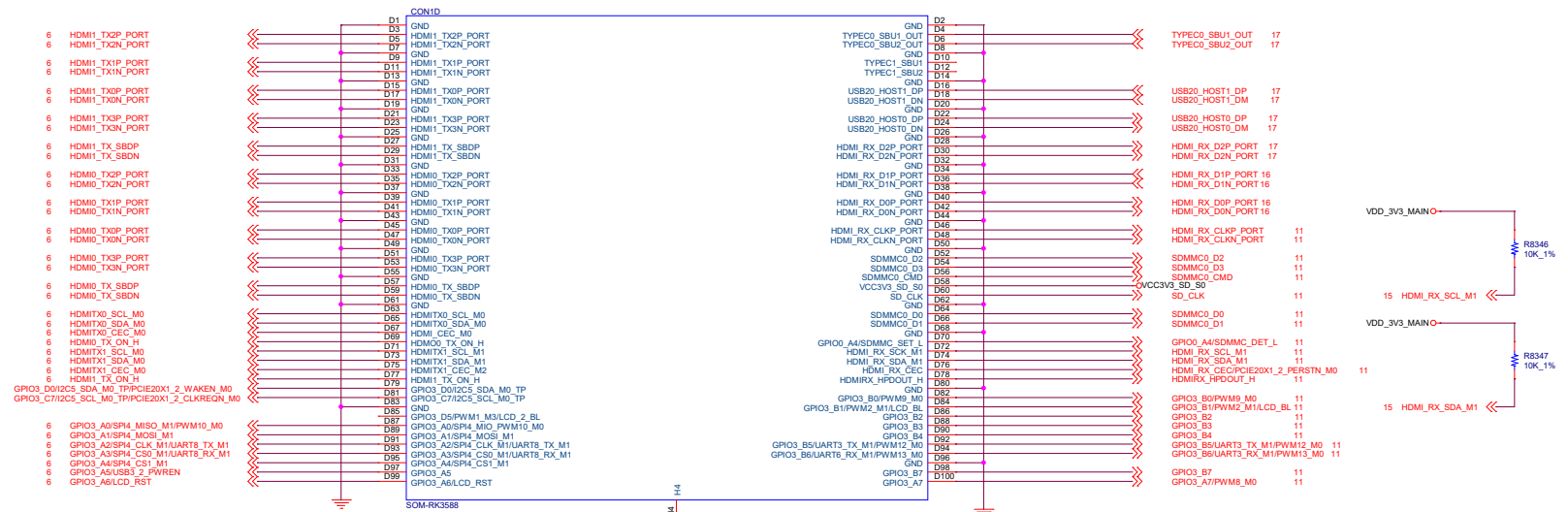
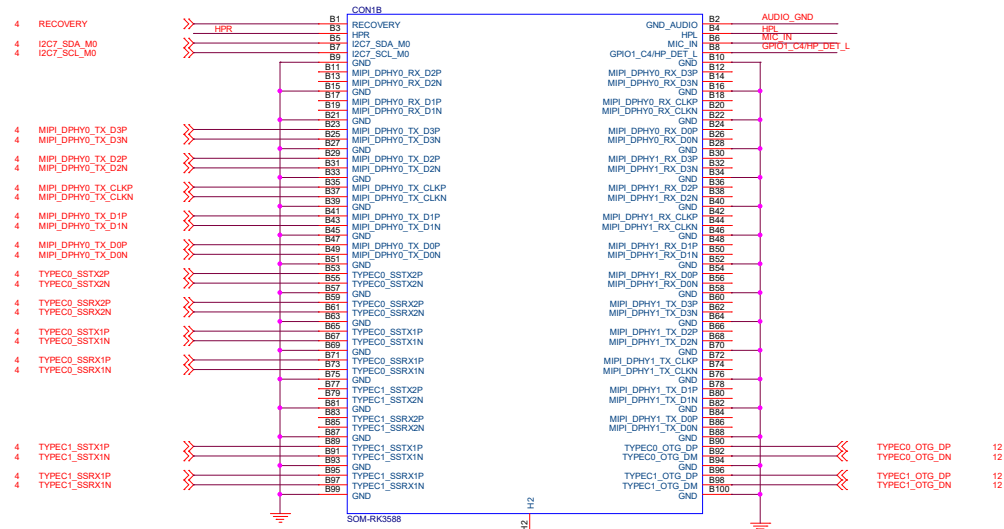
12 GPIO1_B1/SPI0_MISO_M2
12 GPIO1_B2/SPI0_MOSI_M2/UART4_RX_M2
12 GPIO1_B3/SPI0_CLK_M2/UART4_TX_M2
12 GPIO1_B4/SPI0_CS0_M2/UART7_RX_M2

R1115 22K_1% SPI_SDO_SLAVE (2.9)
R1117 22K_1% SPI_SDI_SLAVE (2.9)
R1118 22K_1% SPI_SCK_SLAVE (2.9)
R1119 22K_1% SPI_CS_SLAVE (2.9)



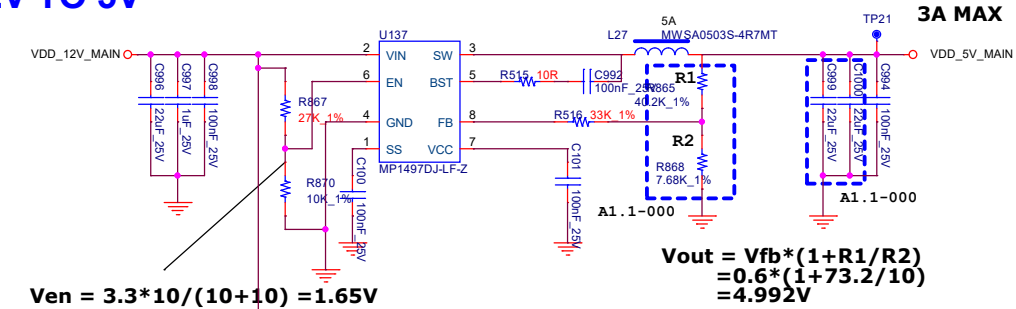
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Size	Document Number	Rev
B	<Doc>	<RevCode>
Date:	Wednesday, May 07, 2025	Sheet 1 of 35

SOM CONNECTOR CON0B/D

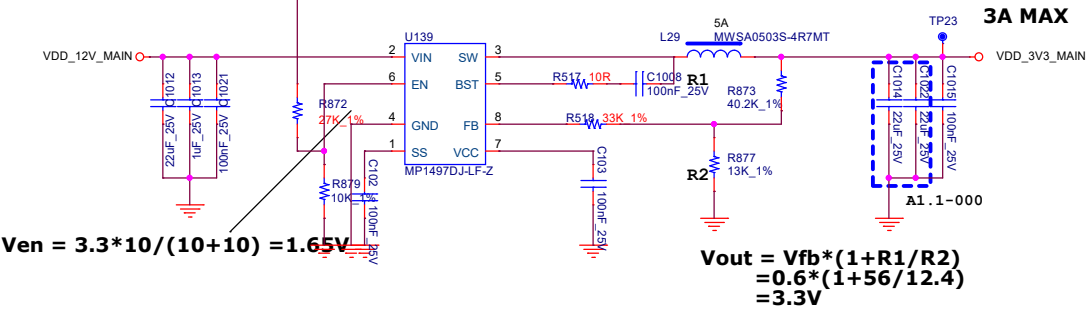


POWER INPUT

12V TO 5V



12V TO 3.3V



Tronlong®

Guangzhou Tronlong Electronic Technology Co., Ltd.

Title

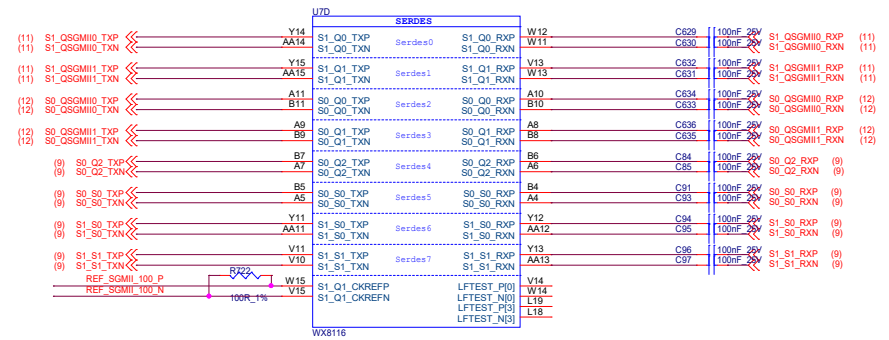
POWER

Size

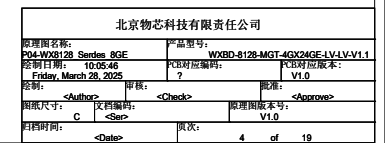
Document Number

Date: Sunday, April 27, 2025

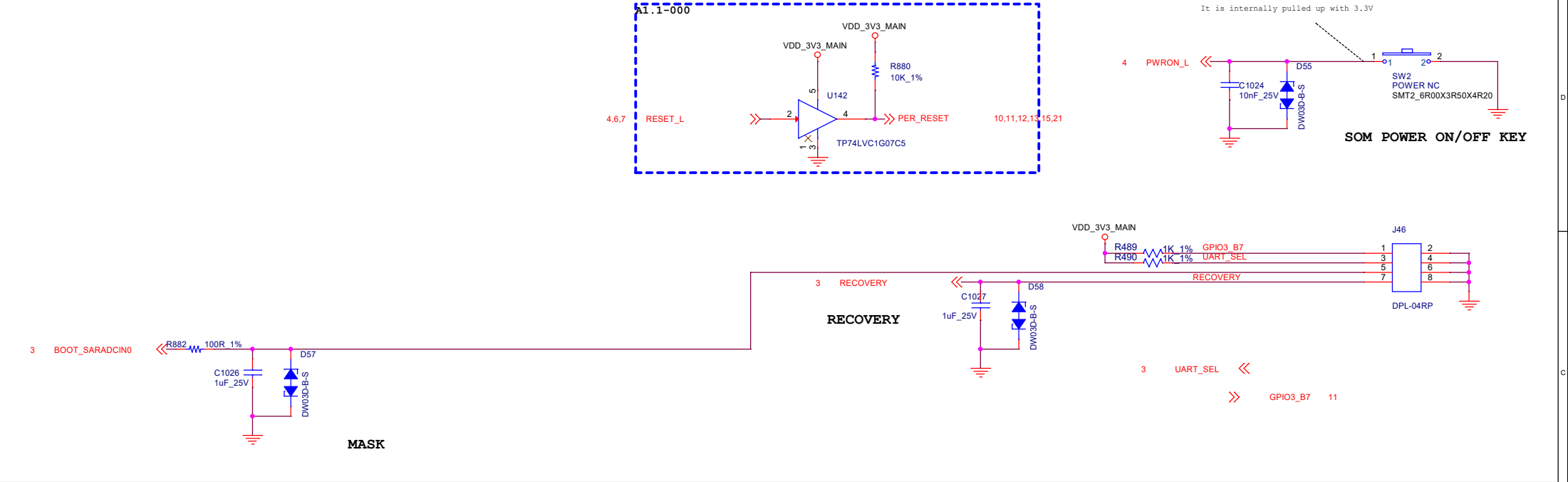
Sheet 3 of 36



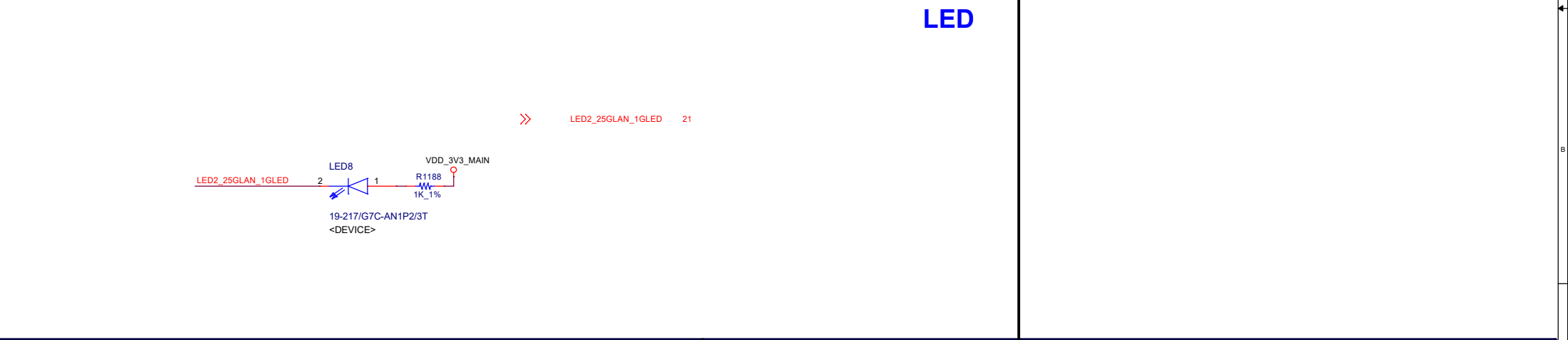
(9)	SFP2_TX+	⋈	S1_S0_TXP		S1_S0_RXP	⋈	SFP2_RX+	(9)
(9)	SFP2_TX-	⋈	S1_S0_TXN		S1_S0_RXN	⋈	SFP2_RX-	(9)
(9)	SFP3_TX+	⋈	S1_S1_TXP		S1_S1_RXP	⋈	SFP3_RX+	(9)
(9)	SFP3_TX-	⋈	S1_S1_TXN		S1_S1_RXN	⋈	SFP3_RX-	(9)



KEY

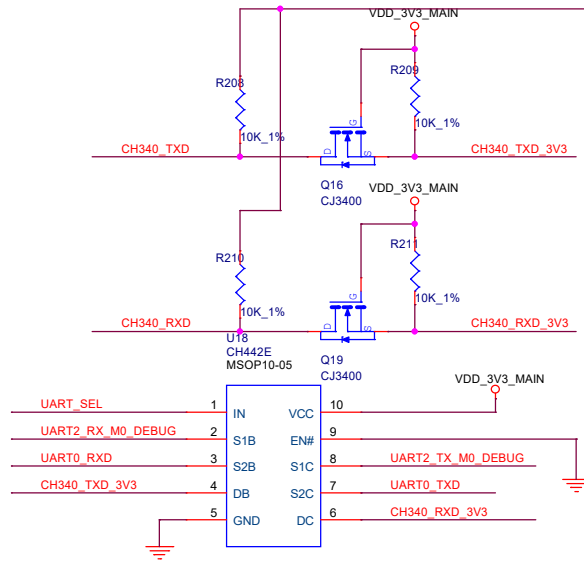


LED

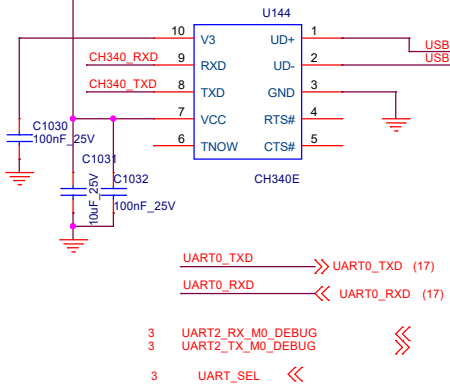


BOOT SET

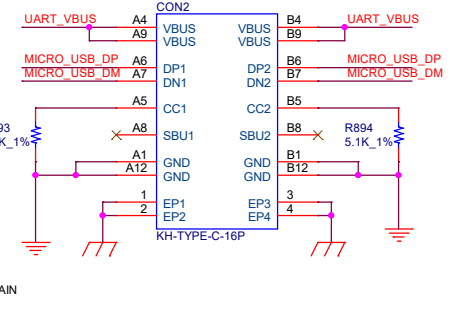
Check the startup program from SPI NOR FLASH, eMMC FLASH, SD Card, and USB PORT in sequence, and start from the first device that has a startup program, no BOOT SET configuration is required



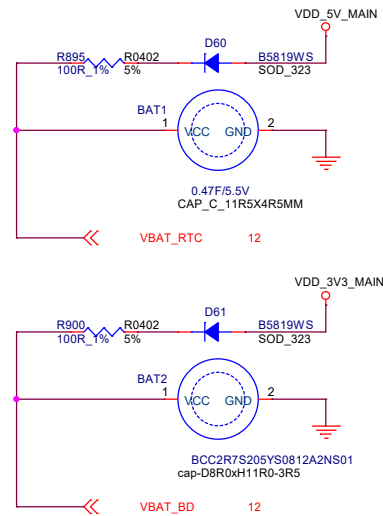
USB TO UART2(DEBUG)



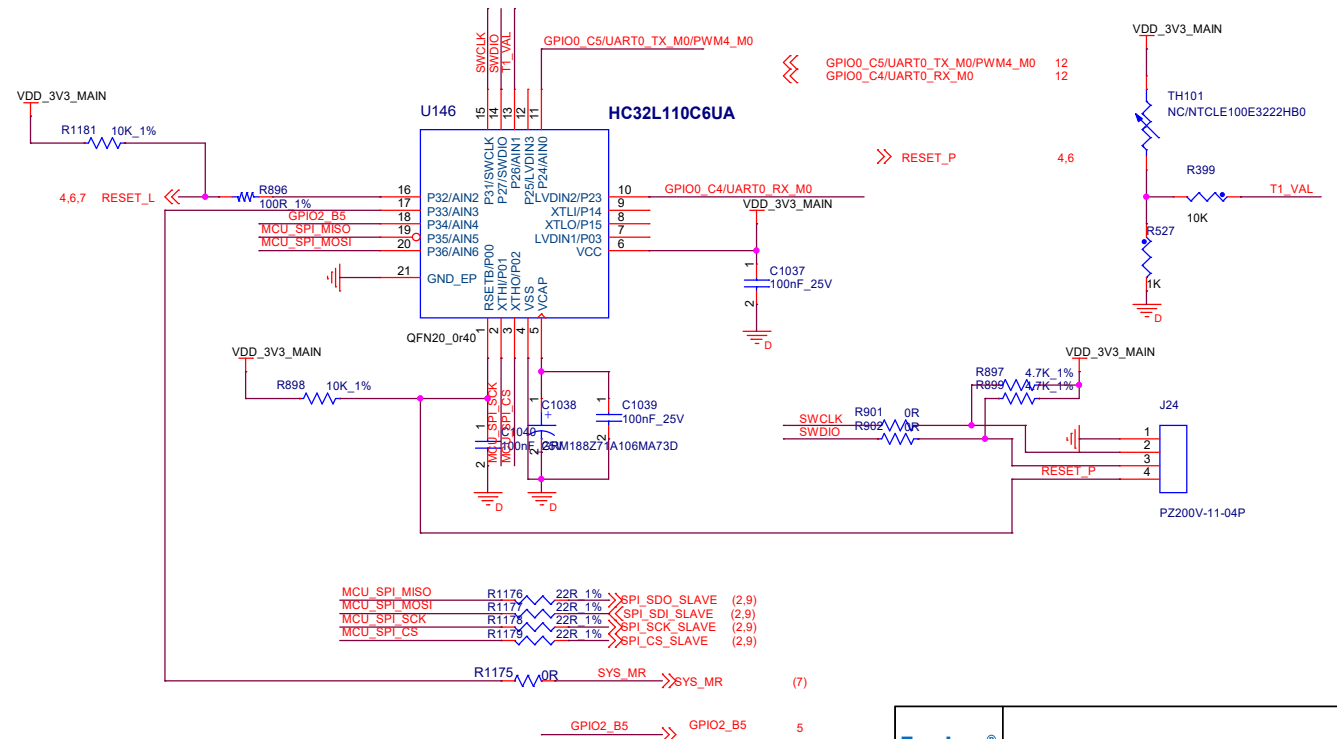
90 Ω (±10%) differential impedance
Keep trace length as short as possible



RTC

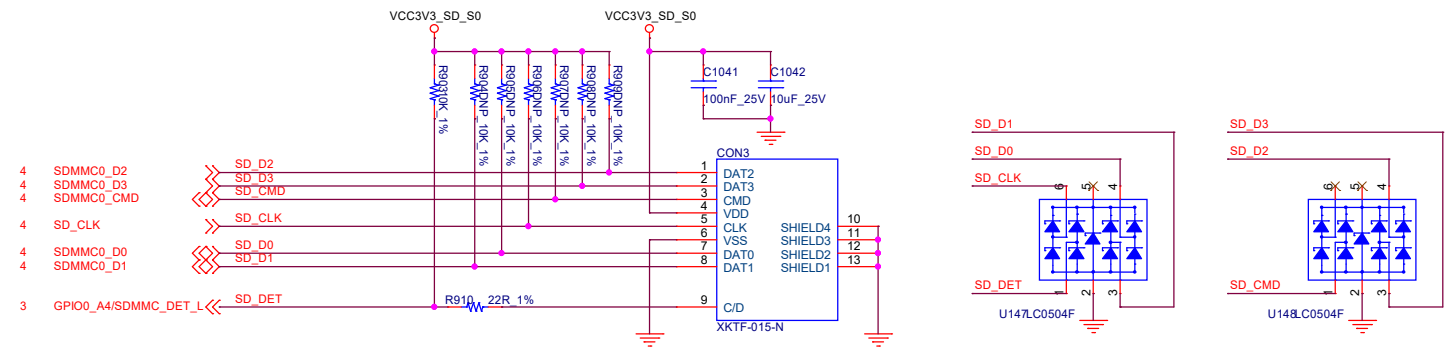


HW WATCHDOG

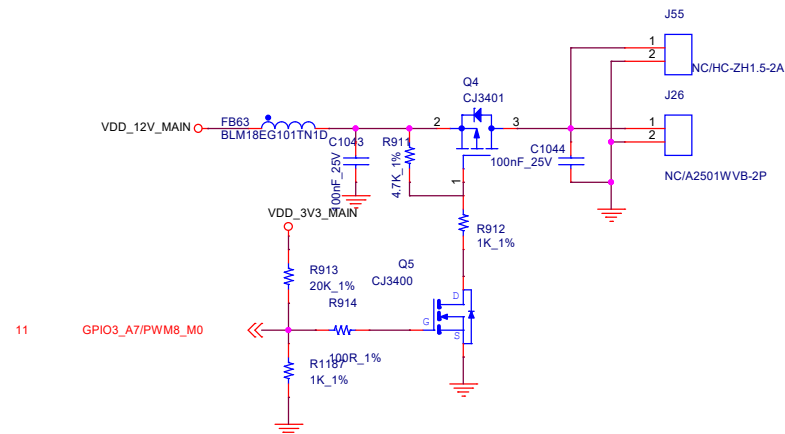


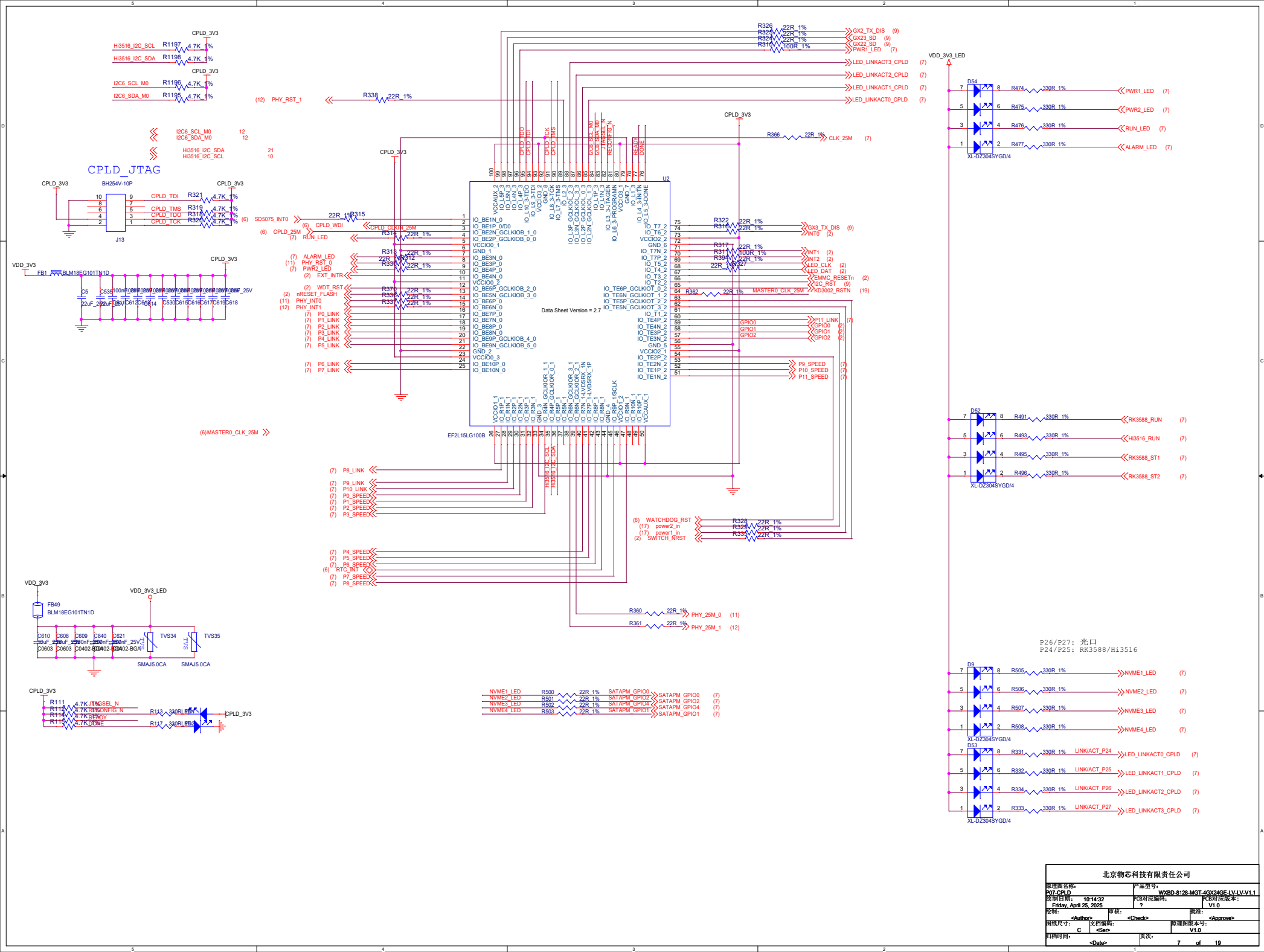
Tronlong®		Guangzhou Tronlong Electronic Technology Co., Ltd.	
Title			
DEBUG UART/RTC/WD			
Size	Document Number		
A3	TL3568-EVM		
Date:	Sunday, April 27, 2025		Sheet 5 of 36

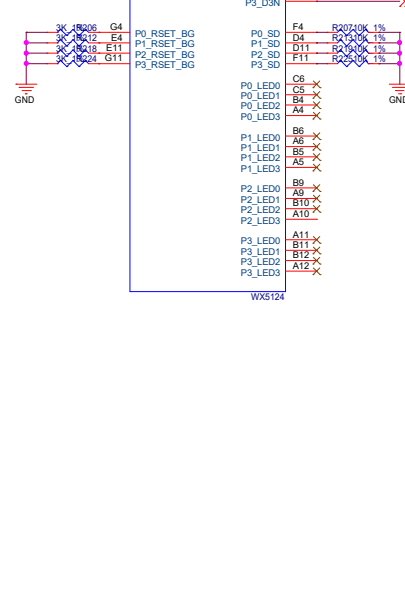
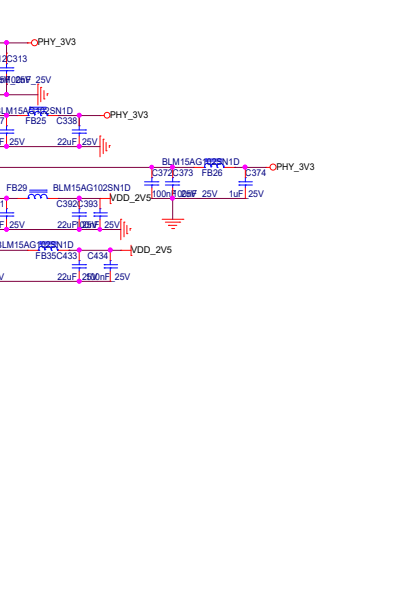
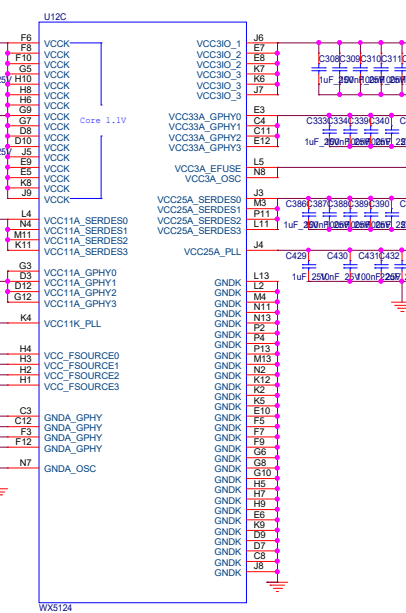
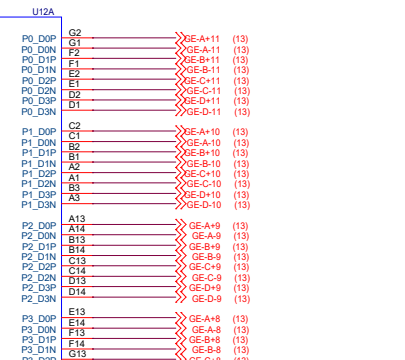
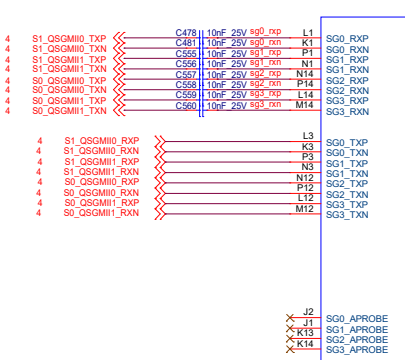
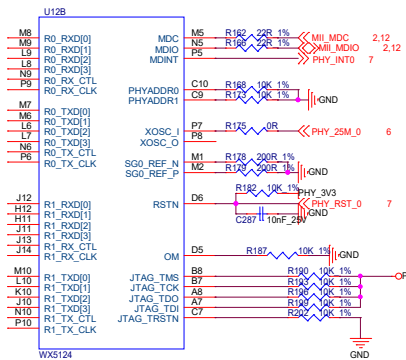
MICRO SD



FAN

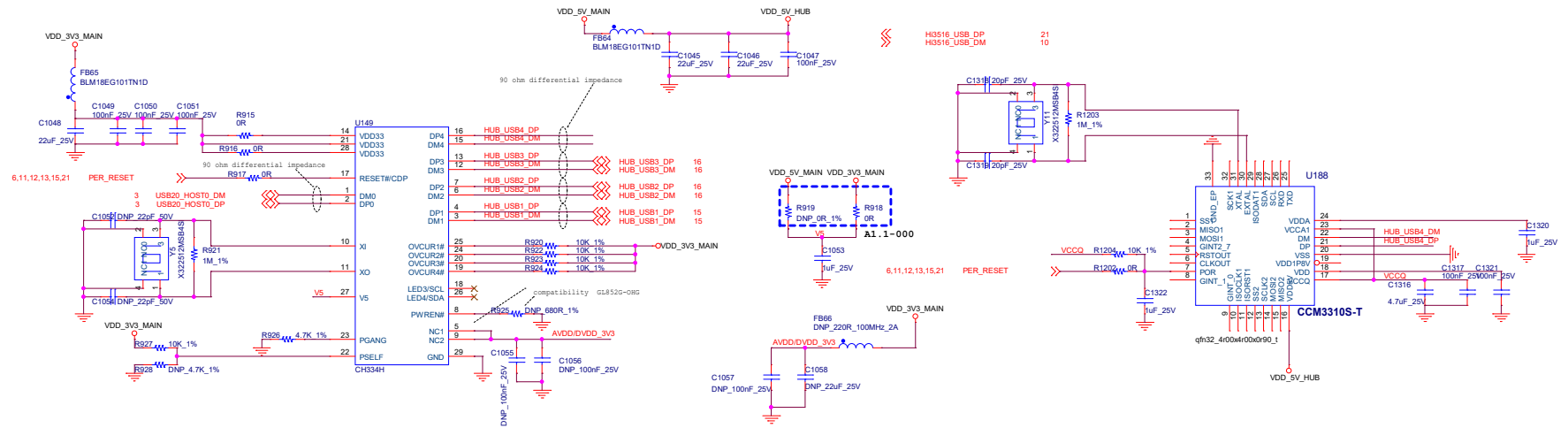




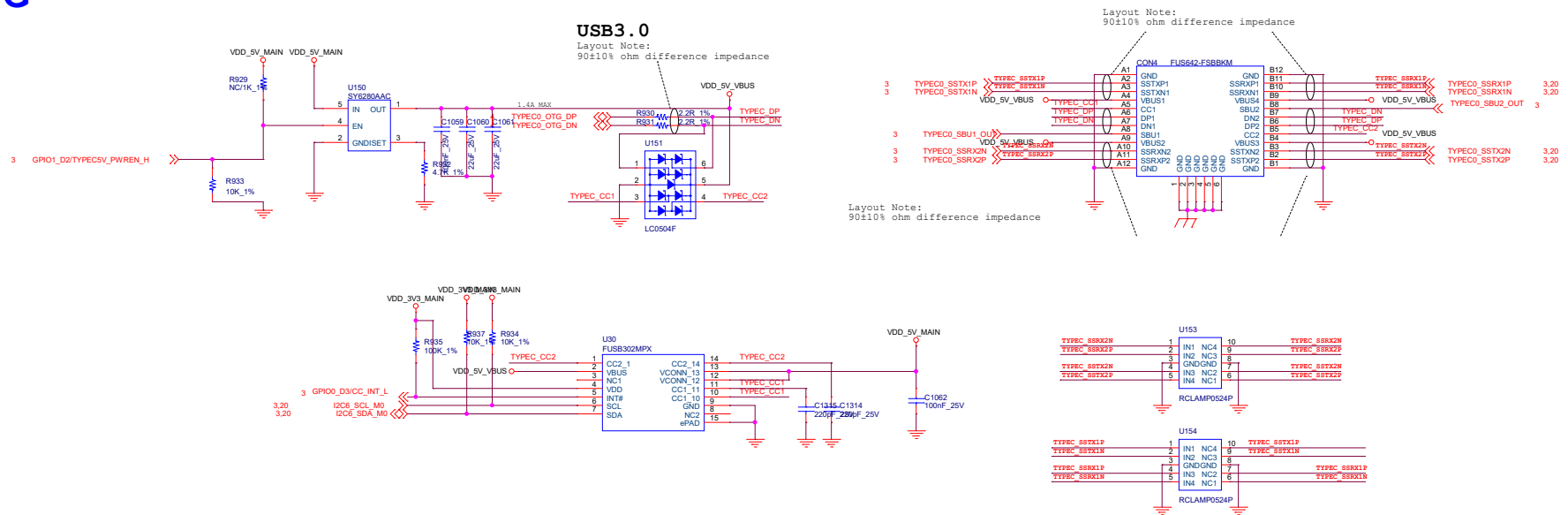


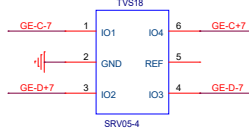
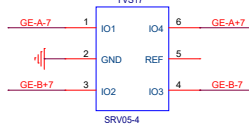
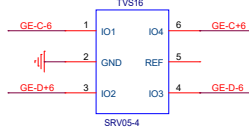
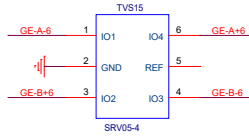
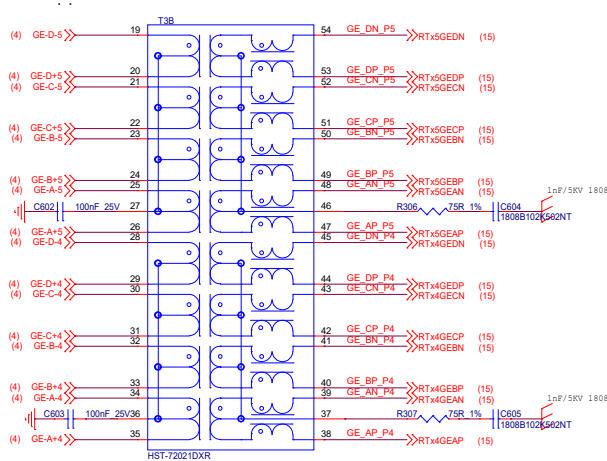
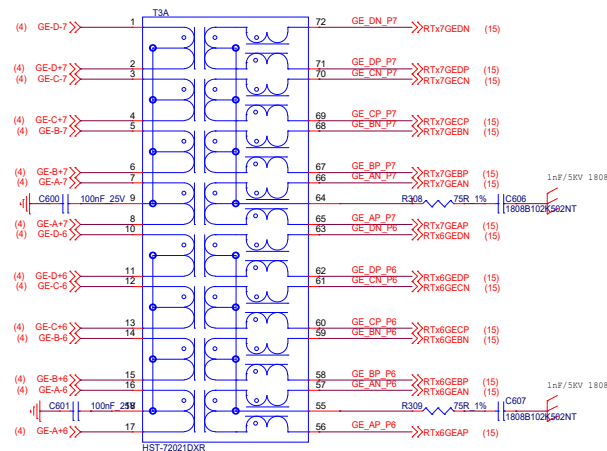
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审核时间:	<Date>	页次:	7 of 16

USB2.0 HUB

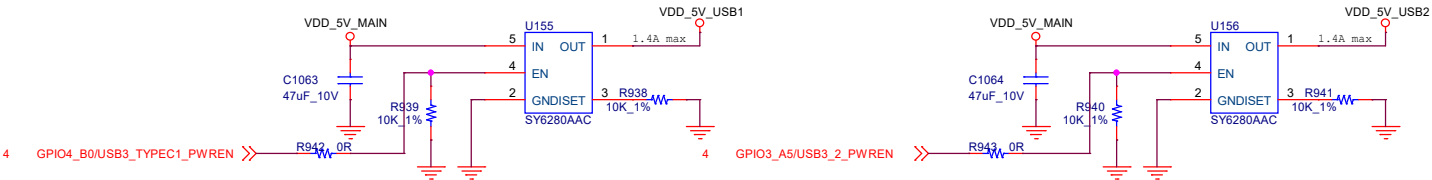


USB3.0 OTG



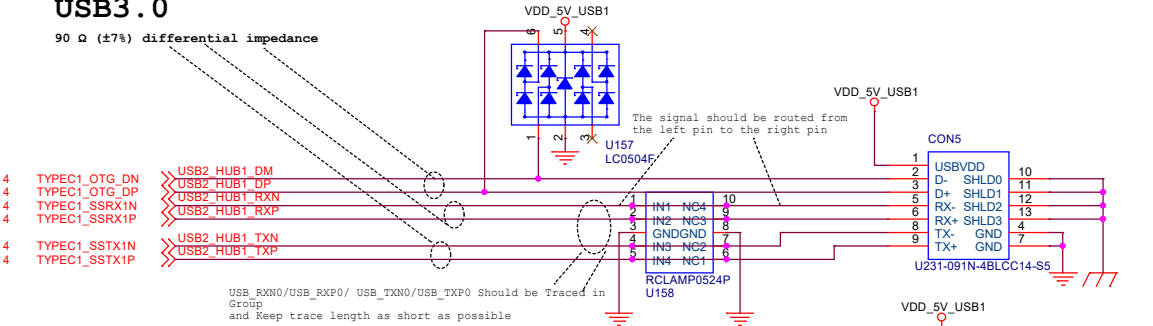


USB 3.0 HOST



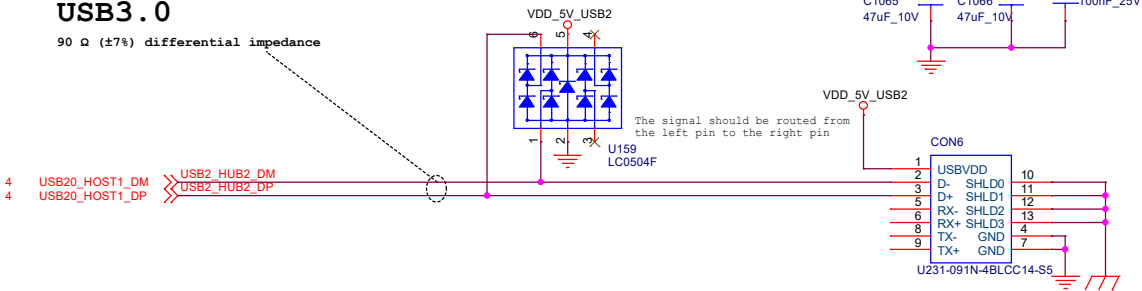
USB3.0

90 Ω (±7%) differential impedance



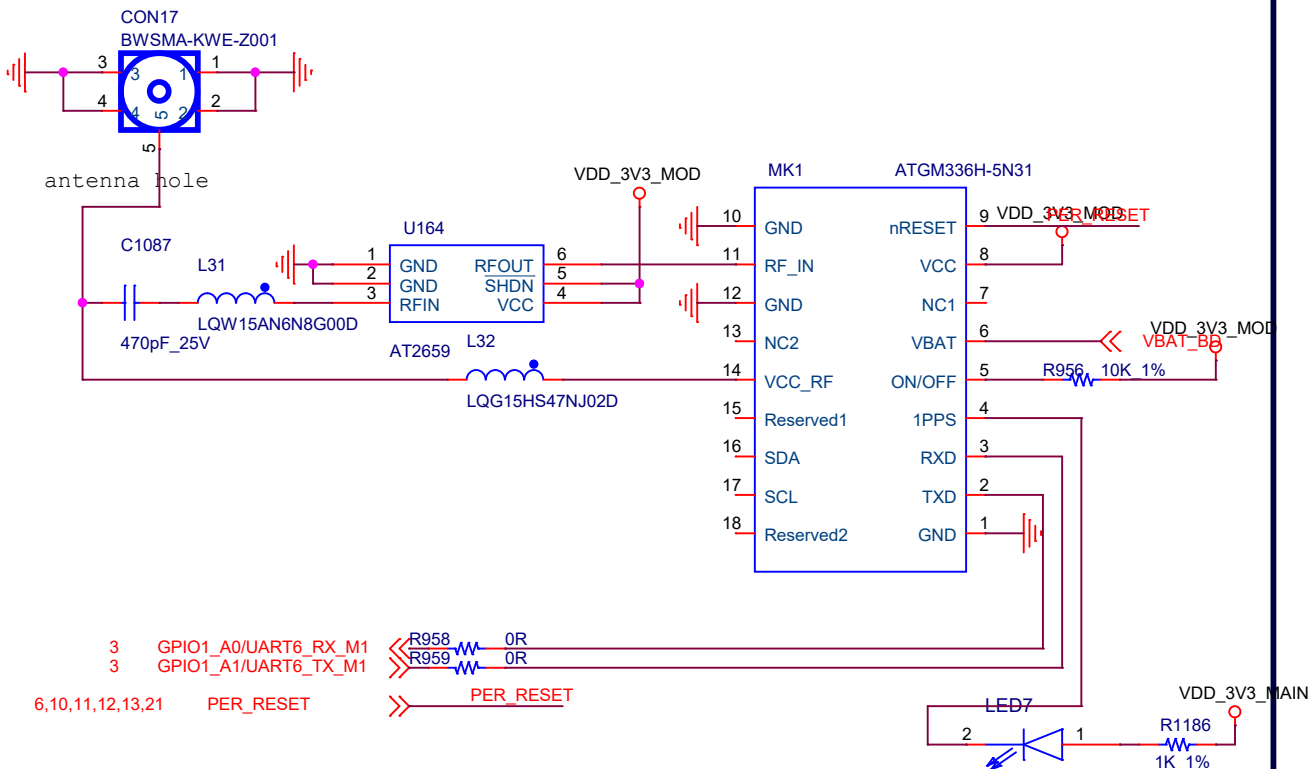
USB3.0

90 Ω (±7%) differential impedance

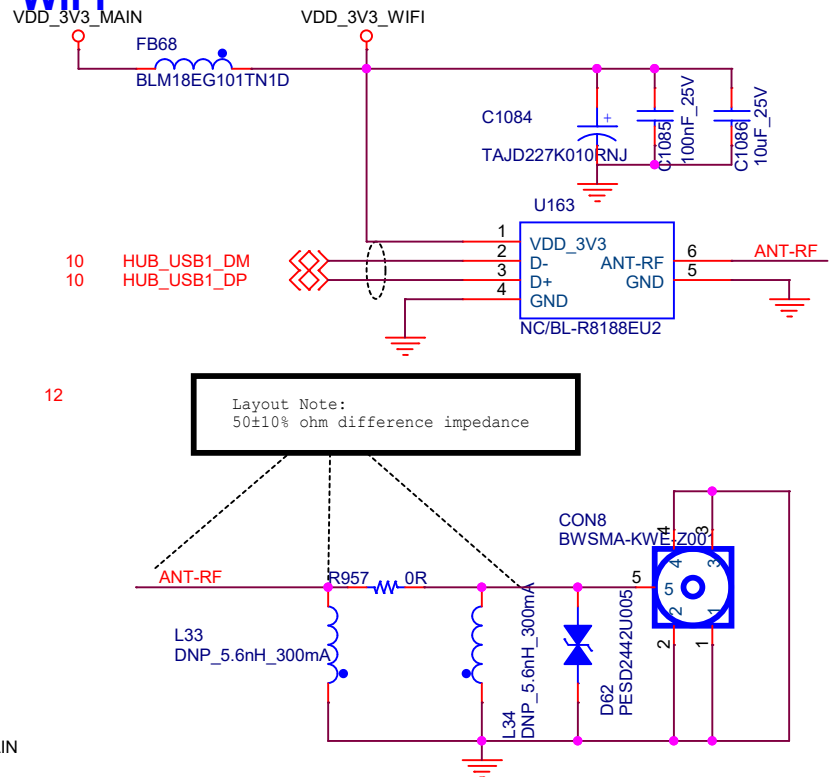




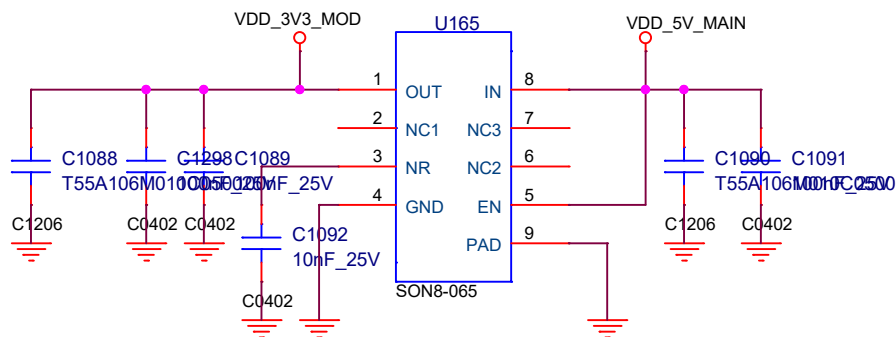
BD/GPS



WIFI

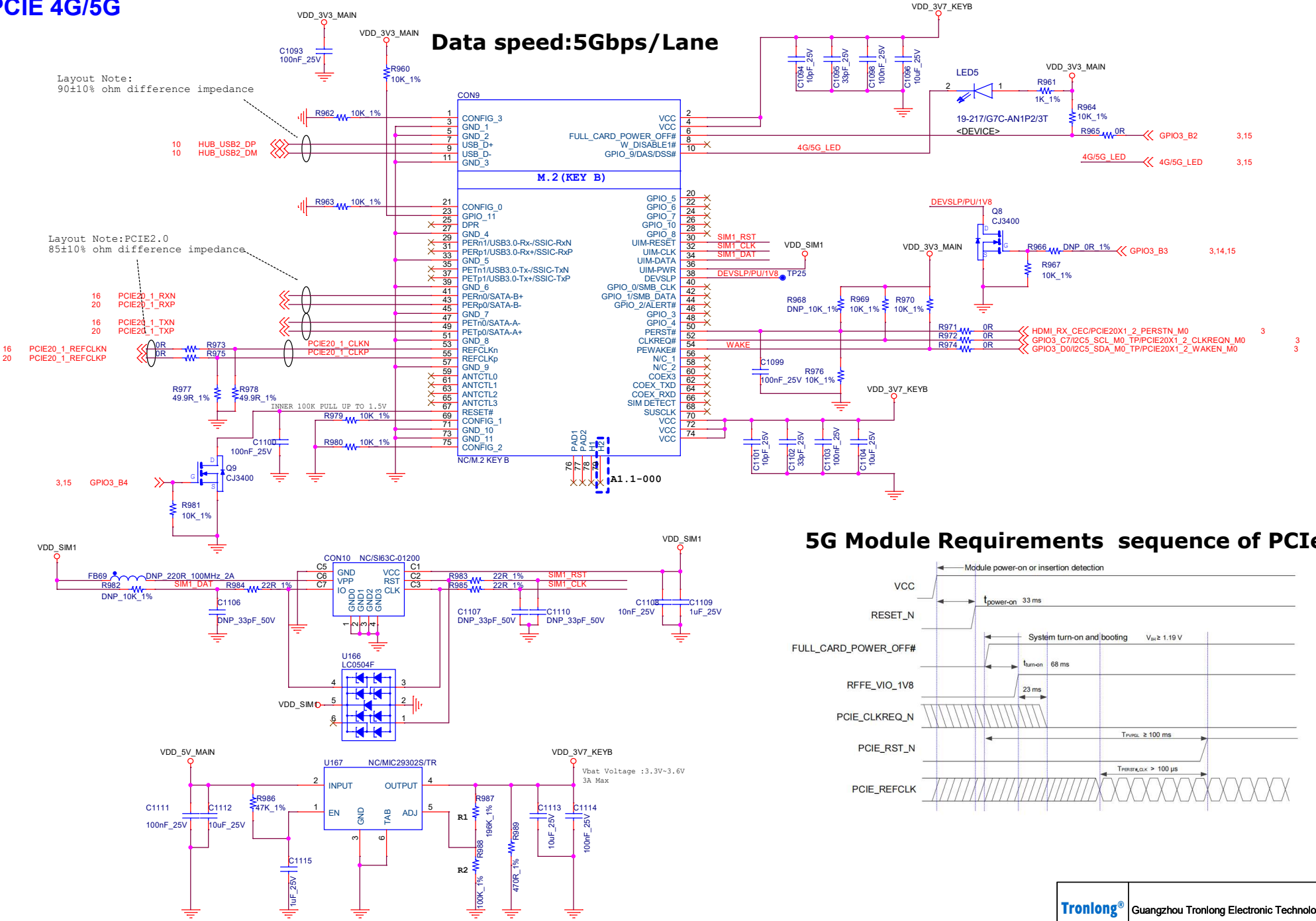


19-217/G7C-AN1P2/3T
<DEVICE>



M.2 PCIE 4G/5G

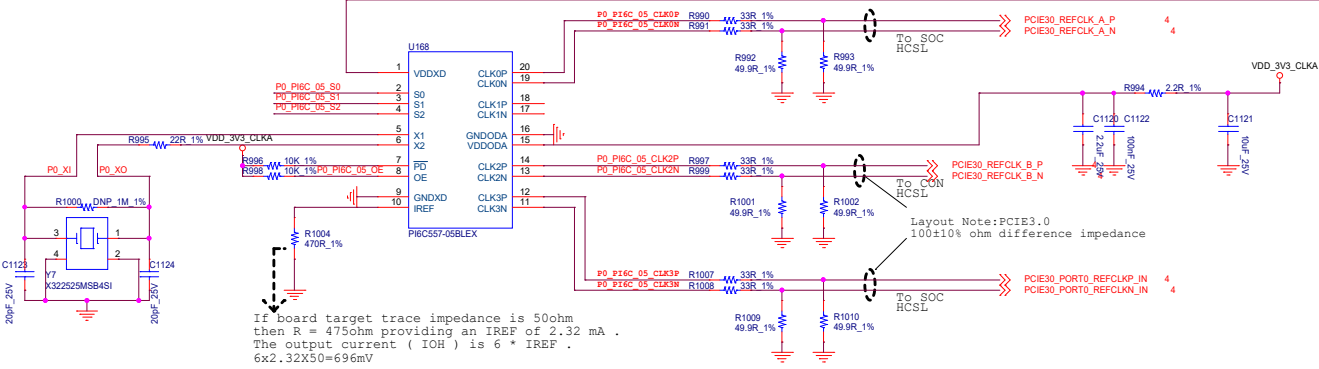
Data speed:5Gbps/Lane



$$V_{out} = V_{fb} \cdot (1 + R1/R2)$$
$$= 1.242 \cdot (1 + 196/100)$$
$$= 3.68V$$

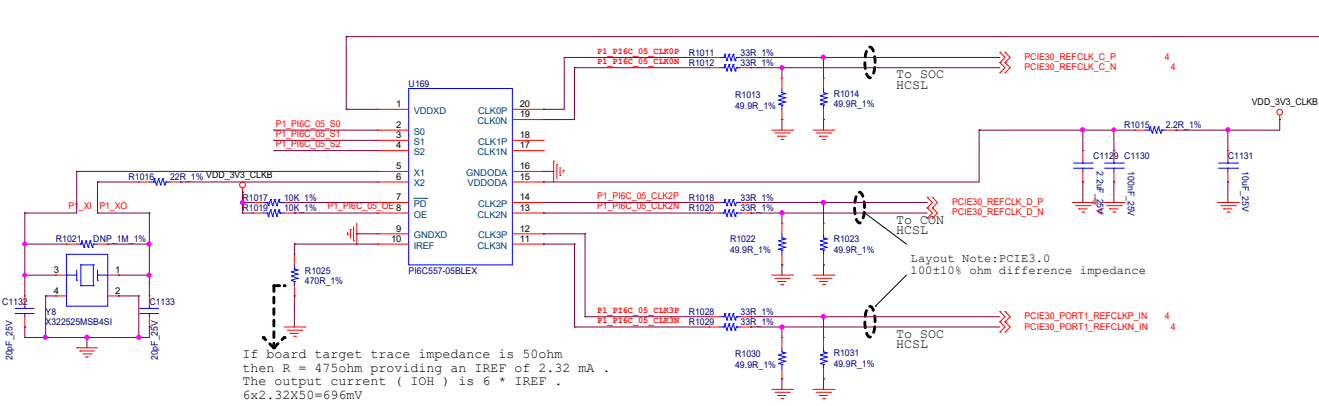
Tronlong®		Guangzhou Tronlong Electronic Technology Co., Ltd.	
Title		M.2 PCIE 4G/5G	
Size		Document Number	
A3		TL3568-EVM	
Date:		Sunday, April 27, 2025	
Sheet		11 of 36	

100MHZ CLK



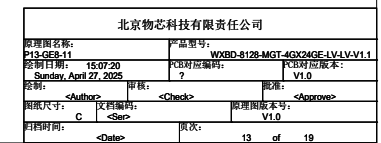
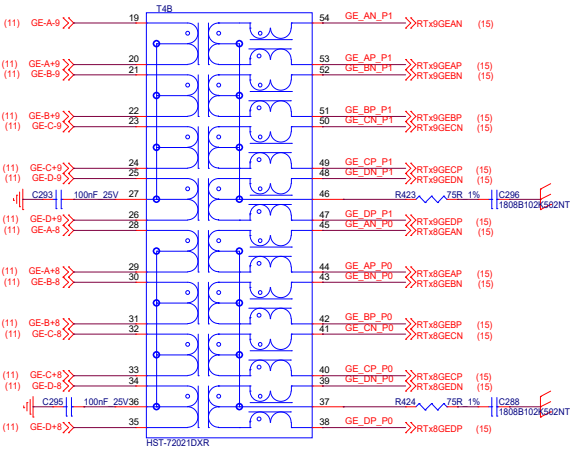
PI6C_S1	PI6C_S0	Out Freq
0	1	100MHz

PI6C_SS1	PI6C_SS0	Spread %
0	0	No Spread
0	1	-0.5
1	0	-1.0
1	1	No Spread

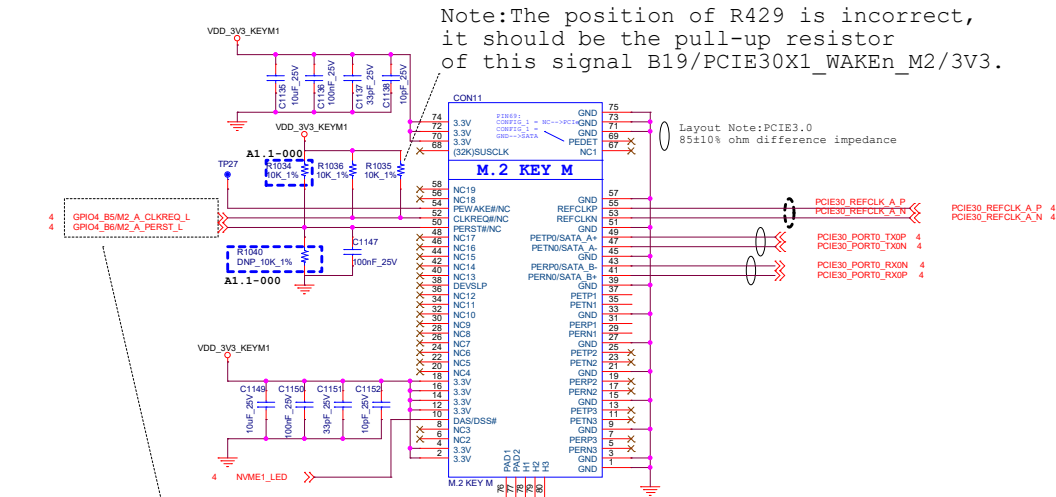


PI6C_S1	PI6C_S0	Out Freq
0	1	100MHz

PI6C_SS1	PI6C_SS0	Spread %
0	0	No Spread
0	1	-0.5
1	0	-1.0
1	1	No Spread



M.2 PCIE NVME



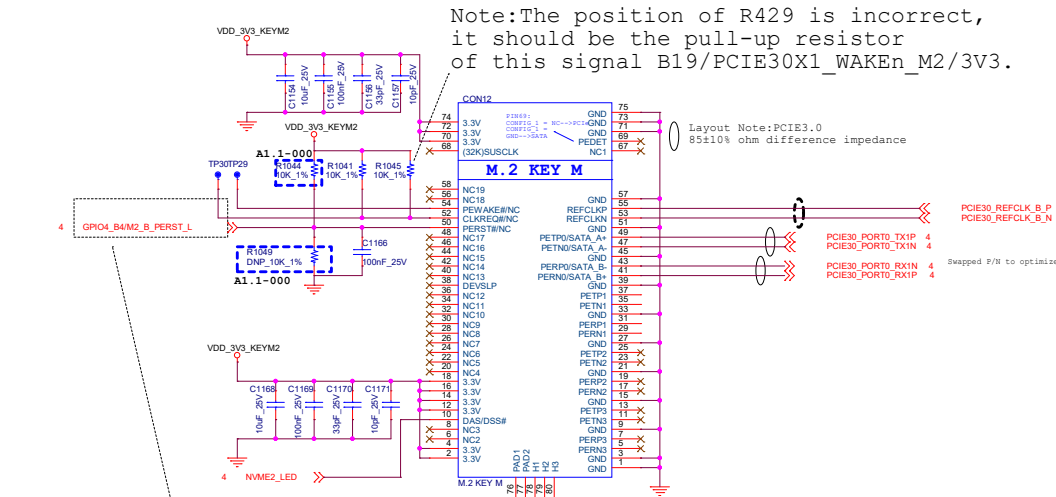
The current three signals design does not meet the PCIE3.0 x2 requirements.

Requirements for the use of this function:

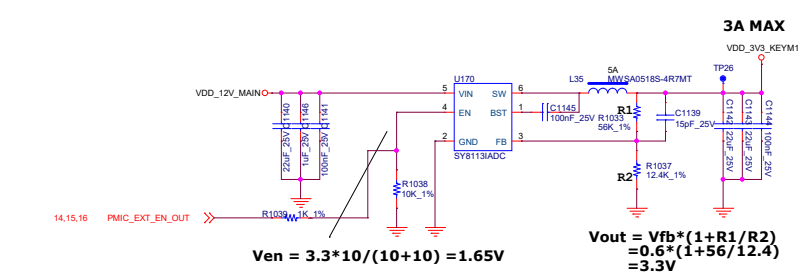
1. PCIE30X2_CLKREQn, PCIE30X2_WAKEn and PCIE30X2_PERSTn are corresponding to Lane PCIE3.0 x2 Dual Mode controller.
2. PCIE30X1_CLKREQn, PCIE30X1_WAKEn and PCIE30X1_PERSTn are corresponding to Lane PCIE3.0 x1 RC Mode controller.

Option1	PCIE3.0 x2Lane	PCIE30_T2D PCIE30_REFCLK (RC/EP input)	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUST00N857n	RC or EP
Option2	PCIE3.0 x1Lane + PCIE3.0 x1Lane	PCIE30_T2D PCIE30_REFCLK (RC input)	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUST00N857n PCIE30X1_CLKREQn PCIE30X1_WAKEn PCIE30X1_PERSTn PCIE30X1_BUST00N857n	Only RC Only RC

M.2 PCIE NVME



12V TO 3.3V



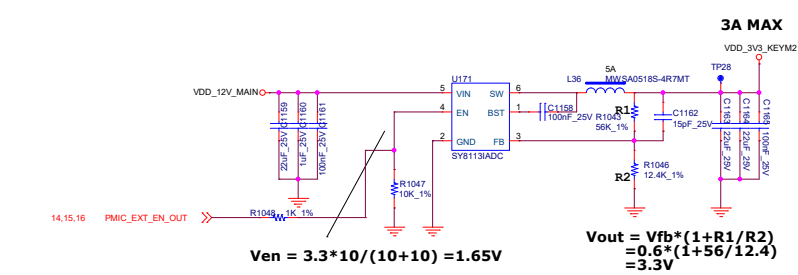
PCie Clock Mode Selection

Mode	Device selection	Description
RC homologous clock	/	Remain the existing design.
RC nonhomologous clock	DNP: R447/R448/R450/R451	
EP homologous clock	DNP: Y8/U80/R447/R448/R450/R451/R454/R455/R456/R457	
EP nonhomologous clock	/	Not support

Note:

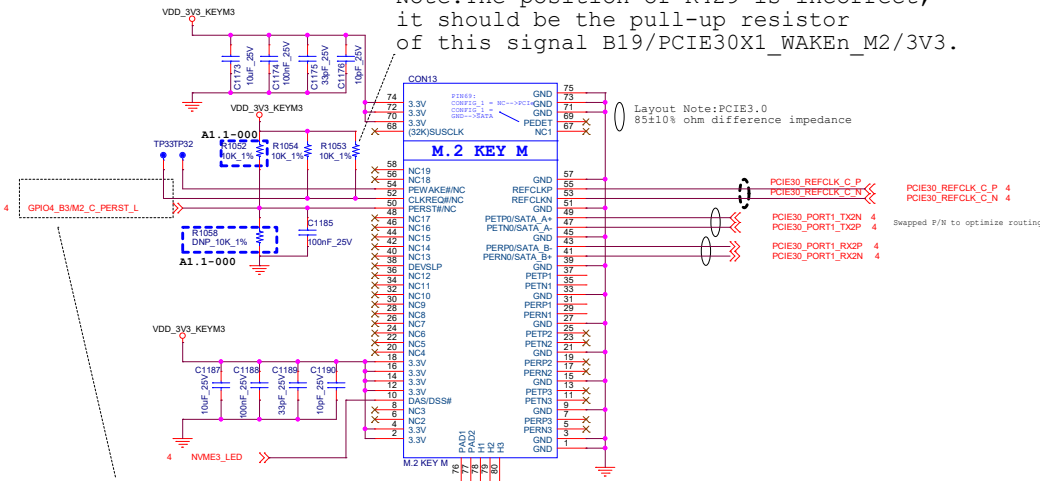
1. PCIE30X2_CLKREQn, PCIE30X1_CLKREQn, PCIE30X2_WAKEn and PCIE30X1_WAKEn must be used as functional pins, and they cannot be replaced by GPIO. Note: select both _M0 or _M1 or _M2 instead of one _M0 and one _M1.
2. PCIE30X2_PERSTn and PCIE30X1_PERSTn can be used as functional pins or be replaced by GPIO. If you choose functional pins, PCIE30X2_PERSTn and PCIE30X1_PERSTn must be in the same Mx group as PCIE30X2_CLKREQn, PCIE30X2_WAKEn and PCIE30X1_CLKREQn, PCIE30X1_WAKEn.

12V TO 3.3V



M.2 PCIE NVME

Note:The position of R429 is incorrect, it should be the pull-up resistor of this signal B19/PCIE30X1_WAKEn_M2/3V3.



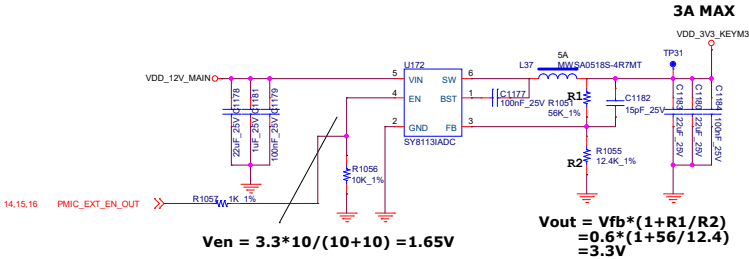
The current three signals design does not meet the PCIE3.0 x2 requirements.

Requirements for the use of this function:

- 1.PCIE30X2_CLKREQn,PCIE30X2_WAKEn and PCIE30X2_PERSTn are corresponding to Lane PCIE3.0 x2 Dual Mode controller.
- 2.PCIE30X1_CLKREQn,PCIE30X1_WAKEn and PCIE30X1_PERSTn are corresponding to Lane PCIE3.0 x1 RC Mode controller.

Option1	PCIE3.0 x2Lane	PCIE30_REFCLK (RC/EP_Lane0)	PCIE30_TX0 PCIE30_RX0 PCIE30_TX1 PCIE30_RX1	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUTTON0nStn	RC or EP
Option2	PCIE3.0 x1Lane + PCIE3.0 x1Lane	PCIE30_REFCLK (RC_Lane0)	PCIE30_TX0 PCIE30_RX0 PCIE30_TX1 PCIE30_RX1	PCIE30X2_CLKREQn PCIE30X2_WAKEn PCIE30X2_PERSTn PCIE30X2_BUTTON0nStn PCIE30X1_CLKREQn PCIE30X1_WAKEn PCIE30X1_PERSTn PCIE30X1_BUTTON0nStn	Only RC Only RC

12V TO 3.3V



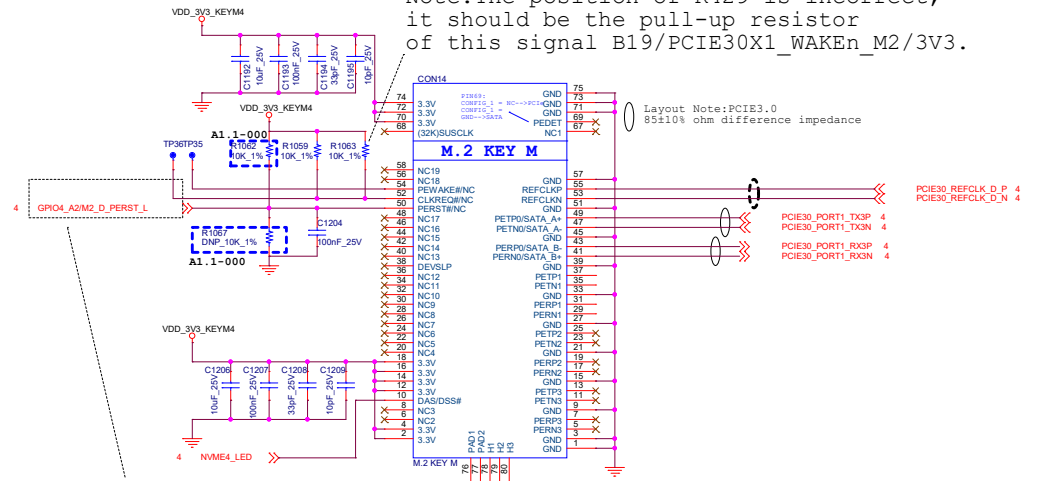
PCie Clock Mode Selection

Mode	Device selection	Description
RC homologous clock	/	Remain the existing design.
RC nonhomologous clock	DNP:R447/R448/R450/R451	
EP homologous clock	DNP:Y8/U80/R447/R448/R450/R451/R454/R455/R456/R457	
EP nonhomologous clock	/	Not support

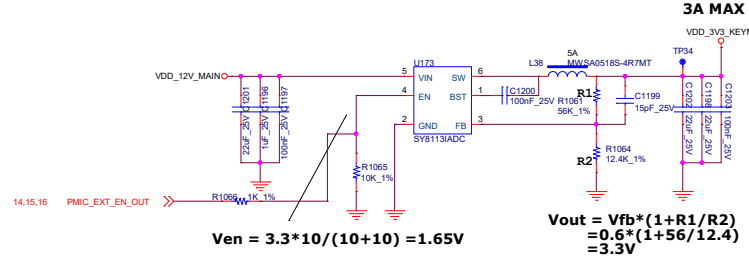
- Note:**
- 1.PCIE30X2_CLKREQn, PCIE30X1_CLKREQn, PCIE30X2_WAKEn and PCIE30X1_WAKEn must be used as functional pins, and they cannot be replaced by GPIO.
 - Note: select both _M0 or _M1 or _M2 instead of one _M0 and one _M1.
 - 2. PCIE30X2_PERSTn and PCIE30X1_PERSTn can be used as functional pins or be replaced by GPIO. If you choose functional pins,PCIE30X2_PERSTn and PCIE30X1_PERSTn must be in the same Mx_group as PCIE30X2_CLKREQn、PCIE30X2_WAKEn and PCIE30X1_CLKREQn、PCIE30X1_WAKEn.

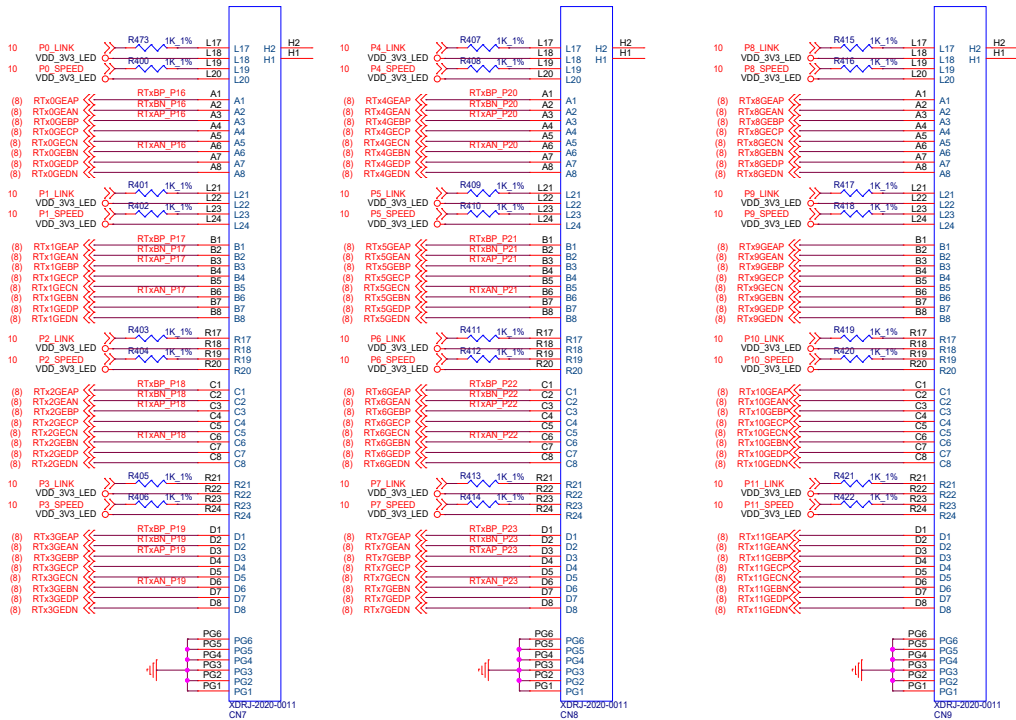
M.2 PCIE NVME

Note:The position of R429 is incorrect, it should be the pull-up resistor of this signal B19/PCIE30X1_WAKEn_M2/3V3.

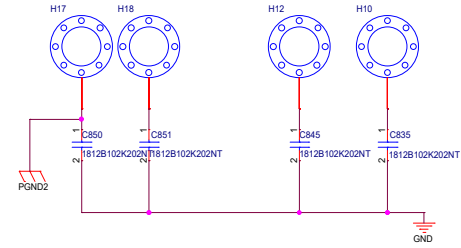
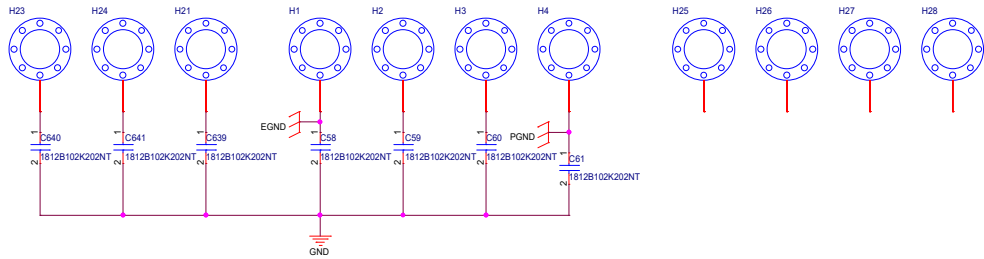


12V TO 3.3V

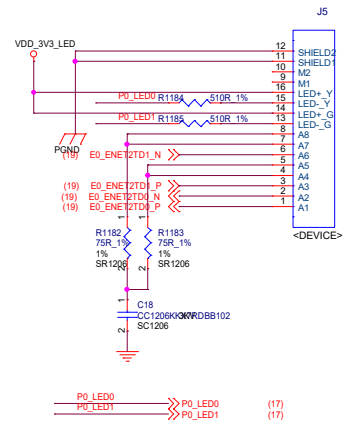
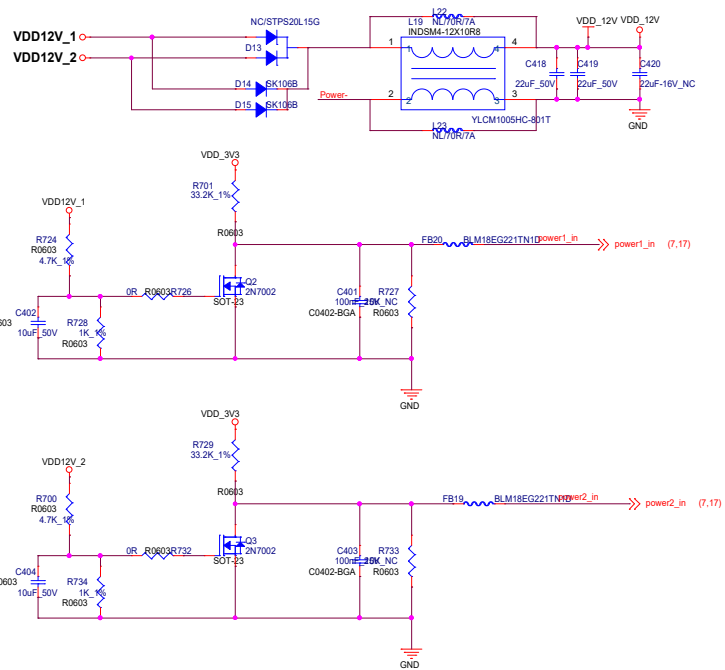
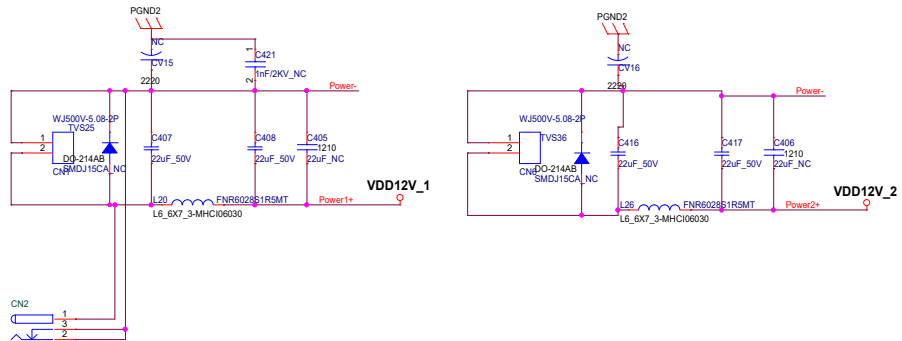




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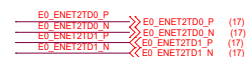


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		页次:	17 of 19

***8128设计注意事项

序号	事项	注意事项
1		每个CPU0可以单独配置成不同模式，Func Mode0【1:0】CPU配置。
2		W3，T6，P7管脚2V5预留
3		GM11不使用时，管脚悬空。 a990C不使用时，管脚悬空。
4		使用芯片内部Q8GM11转4*8GM11时，只能用S1_Q1。 8128上电时序，先3.3V再1.1V

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原理图名称:		产品型号:	
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归档时间:		页次:	
<Date>		18	of 19



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产品名称: PH-PORT3002 控制日期: 11:20:43 Saturday, April 19, 2025		产品型号: WXB-D128-MGT-40X34-LV4-V.1.1 PCB控制编码: ? 固件版本号: V1.0	
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